Biliary Atresia, Part Two: Pathogenesis. Viral and Immune Hypotheses, Biliatresone, and a Disease Present at Birth.

Imagine a mother sitting across from you two days after her daughter's Kasai operation. The surgeon has confirmed biliary atresia. She asks the question every parent asks: what caused this, and was it something I did? A second question comes from the surgical team the same afternoon: should we give steroids, since the disease is inflammatory? And a third comes from the laboratory, which reports that the baby's serum cytomegalovirus IgM is positive. Was the virus the cause? To answer all three questions well, you need to understand what is known, and what is not known, about how biliary atresia begins.

The clinical question of this chapter is therefore this. Who strikes the bile duct, and when? The main idea is that biliary atresia is best understood as a disease of several hits rather than a single cause. First there is a susceptible, developing extrahepatic bile duct, sometimes carrying genetic variants that weaken it. Second there is an insult to the lining cells of the duct, the cholangiocytes, which may be a virus or a toxin and which usually happens before birth or in the first weeks of life. Third there is an immune response that amplifies the injury, closes the lumen of the duct and drives scarring. The last step is fibrosis: the duct becomes a fibrous cord and the liver begins to scar. No single hypothesis explains every infant, and the different phenotypes described in the previous chapter probably reflect different combinations of these hits.

A note on sources. The guideline framework is the 2017 joint guideline on cholestatic jaundice in infants by Fawaz and colleagues, written for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. The timing evidence comes from Sanjiv Harpavat and colleagues at Texas Children's Hospital, in a 2011 study in Pediatrics and a 2020 newborn screening study in the Journal of the American Medical Association. The immune evidence comes largely from the laboratories of Jorge Bezerra in Cincinnati and Cara Mack in Colorado, and from a 2020 single-cell study of human livers published in Cell by Wang and colleagues in China. The trial evidence comes from the START trial of steroids, published in 2014, the King's College Hospital steroid trial of 2007, a 2018 Cochrane review, and the PRIME trial of intravenous immunoglobulin, published in 2019. The toxin story comes from Lorent and colleagues in 2015 and Waisbourd-Zinman and colleagues in 2016, both from the group of Rebecca Wells and Michael Pack in Philadelphia, and from a 2024 human organoid study from Hong Kong. Neither the American Association for the Study of Liver Diseases, known as AASLD, nor the European Association for the Study of the Liver, known as EASL, has a guideline that addresses the pathogenesis of biliary

atresia.

Part one. A cascade of several hits.

It helps to picture five steps in a row. The first step is susceptibility. Some infants carry rare variants in genes that build cilia, the tiny antenna-like structures on the surface of cholangiocytes, or genes that set the left and right sides of the body. Others carry common variants that slightly raise risk. And all infants share one structural weakness: the extrahepatic bile duct contains less glutathione, the main antioxidant inside cells, than the liver itself, which makes it unusually vulnerable to chemical injury.

The second step is a trigger. This could be a virus, such as cytomegalovirus or reovirus, or a reactive chemical toxin, and it reaches the cholangiocyte before birth or in the first weeks after it.

The third step is epithelial injury. In experimental systems, glutathione falls, a transcription factor called SOX17 that maintains the duct falls with it, cilia are lost, and the tight junctions that seal the spaces between cholangiocytes come apart. The duct starts to leak, and its lumen collapses.

The fourth step is immune amplification. Type one helper T cells, which produce the cytokine interferon gamma, natural killer cells, and B cells that make antibodies all attack the injured duct, while regulatory T cells, the cells that normally restrain an immune response, fail to apply the brake.

The fifth step is obliteration and fibrosis. The duct becomes a fibrous remnant, small new bile ductules proliferate at the edge of the portal tracts, which pathologists call a ductular reaction, and fibrosis spreads through the liver. An enzyme called matrix metalloproteinase seven, abbreviated MMP-7, leaks from the injured lining cells into the blood.

Two practical messages follow from this cascade. In mouse models, switching off any single step, whether interferon gamma, natural killer cells, B cells, or glutathione loss, can prevent atresia. But in human infants, the cascade is already far advanced by the time jaundice is noticed. That is the central problem for any therapy given after the Kasai operation. And because MMP-7 reads out the third step, it has become a diagnostic test, while a drug aimed at the third step, N-acetylcysteine, is now being tested in a trial.

Part two. The injury starts before the first feed.

The most important constraint on any theory of pathogenesis is timing. In 2011, Harpavat, Finegold and Karpen looked back at sixty-one infants with biliary atresia treated at Texas Children's Hospital and found that roughly half had had a fractionated bilirubin measured in the first days of life, usually as part of routine newborn jaundice checks. In those infants, the

mean direct bilirubin was about 1.4 milligrams per decilitre, compared with about 0.19 milligrams per decilitre in control newborns, a highly significant difference. In other words, the direct bilirubin was already raised seven-fold before any parent or clinician had noticed jaundice or pale stools.

There was a subtle trap in the same data. Most of these infants had a ratio of direct to total bilirubin of 0.2 or less, which is the threshold many clinicians use to decide that jaundice is not cholestatic. In the first days of life, total bilirubin is high for physiological reasons, so the ratio looks normal even when the direct bilirubin is clearly abnormal. The lesson is that the absolute direct or conjugated value matters, not the fraction. In a follow-up series from the same group, every one of thirty-five infants with atresia had a raised newborn direct bilirubin, while 98.2 percent of infants without atresia had normal values.

Why does this matter for pathogenesis? Because conjugated bilirubin retention within a day or two of birth means that the obstruction, or at least the cholangiocyte injury that precedes it, was already present before birth. That argues strongly against a purely postnatal viral trigger. It also fits the phenotypes we already know to be prenatal: the cystic form, which can be seen on the anomaly scan at sixteen to twenty weeks, and the splenic malformation form, whose associated anomalies arise in the fourth to sixth week of embryonic life. The joint NASPGHAN and ESPGHAN guideline says the same thing in plain words: the biliary injury begins before, or soon after, birth. The guideline also defines any direct or conjugated bilirubin above 1.0 milligram per decilitre, which is seventeen micromoles per litre, as pathological.

Part three. If it is there at birth, screen at birth.

If the disease is biochemically visible at birth, it should be possible to find it at birth. In 2020, Harpavat and colleagues reported a two-stage screening programme across fourteen Texas hospitals between 2015 and 2018. At the first stage, within the first sixty hours of life, a direct or conjugated bilirubin was measured, and a value above the hospital's newborn reference range, for conjugated bilirubin above 0.2 milligrams per decilitre, counted as positive. At the second stage, by the two-week visit, the test was repeated, and it counted as positive if it was higher than the first value or above 1 milligram per decilitre.

The numbers were striking. Of 124,385 newborns screened, 119 remained positive at the second stage. Of those, 112 were false positives and seven had biliary atresia, a birth prevalence of 0.6 per ten thousand. If you picture these results as bars, the screened population is enormous, the second-stage positives are a small group, and the atresia cases are a tiny sliver. Sensitivity was 100 percent, with a ninety-five percent confidence interval from 56 to 100 percent, specificity was 99.9 percent, and the positive predictive value was 5.9 percent. The authors were careful to note that they could not fully ascertain false negatives, so 100 percent sensitivity is an upper bound rather than a guarantee. The clinical payoff was clear: the mean age at the Kasai operation fell from 56 days in the historical period to 36 days after

screening began, nineteen days earlier. A single-centre implementation study published in 2025 by Rabbani and Shah found that 1.3 percent of babies were flagged at first, but the net false positive rate after repeat testing was only 0.3 percent.

Part four. The viral hypothesis: a virus at the scene is not a culprit.

The oldest theory of biliary atresia is that a virus infects the duct. Several viruses have been found in the livers or duct remnants of affected infants. Cytomegalovirus, as IgM antibody in serum or as viral DNA in tissue, is found in roughly ten to thirty-eight percent of infants in European and Brazilian series, and in up to sixty percent in some Chinese series. In 1998, Tyler and colleagues reported reovirus RNA in hepatobiliary tissue from infants with atresia, but also from infants with choledochal cysts, so the finding was not specific. Human papillomavirus has not been found, and data for human herpesvirus six and Epstein-Barr virus are inconsistent. Characteristic cytomegalovirus inclusion bodies have never been demonstrated in atresia livers, and recent reviews conclude that no virus has been confirmed as causal.

The most useful contribution of virology has been an animal model. In 1993, Riepenhoff-Talty and colleagues showed that rhesus rotavirus given to newborn mice produces obstruction of the extrahepatic bile ducts, portal fibrosis and ductular proliferation. This model became the workhorse of the field, and most of what we know about the immune biology of atresia comes from it. It shows how a virus can start an immune attack on a duct that is still maturing in the newborn period. But a mouse model is not proof that rotavirus causes the human disease, and it does not.

The deeper problem with the viral hypothesis is timing and sampling. Detection depends on the infant's age when the sample is taken, on the assay used, and on geography. A virus acquired after birth can colonise a liver that was already injured, so finding it at the time of the Kasai cannot tell you when the injury began.

Part five. Cytomegalovirus rises with age at surgery.

The best illustration of this problem comes from a 2022 study by Fischler and colleagues pooling four European centres, in Budapest, London, Stockholm and Warsaw. Overall, about one in five infants with atresia, eighty-one of 407, was cytomegalovirus positive at the time of the Kasai. But positivity climbed steeply with age. None of fourteen infants operated before thirty days of age was positive. Seventeen percent of those operated between thirty-one and seventy days, fifty-three of 308, were positive. And thirty-three percent of those operated after seventy days, seventy-eight of 234, were positive. Drawn as a bar chart, the bars rise from nothing to a sixth to a third.

That gradient fits postnatal acquisition: the longer the infant lives before surgery, the more chance it has had to pick up a common virus. Yet cytomegalovirus is not a harmless passenger. In London, cytomegalovirus-positive infants cleared their jaundice after the Kasai much less often, thirty-seven percent compared with seventy-one percent. In a 2025 Chinese multicentre study of 1,276 infants with type three atresia, 31.3 percent were cytomegalovirus positive, and they were older at surgery, a median of 68 days compared with 57. Antiviral treatment in that study reduced early cholangitis, from fifty percent to about ten percent, but did not improve jaundice clearance or one-year native liver survival. And in 2012, Brindley, Mack and colleagues in Colorado showed that in fifty-six percent of infants with atresia, T cells from the liver produced interferon gamma when exposed to cytomegalovirus proteins, and that regulatory T cells in the blood were reduced, most markedly in cytomegalovirus-positive infants. So cytomegalovirus may act as an accelerator of immune injury in a subgroup, whether or not it started the disease.

Part six. The immune hypothesis: switch off one arm, prevent atresia.

The rhesus rotavirus model allowed researchers to ask which parts of the immune system are necessary for atresia. The answers were remarkably consistent. In 2004, Shivakumar, Bezerra and colleagues showed that in ordinary mice given the virus, persistent cholestasis developed in about eighty percent, and fewer than ten percent survived beyond twenty-one days. In mice lacking interferon gamma, jaundice appeared at first but then resolved, and more than eighty percent survived. Giving recombinant interferon gamma back to these mice restored cholestasis in more than eighty percent. That is strong evidence that the type one helper T cell pathway drives duct obstruction.

In 2009, the same group showed that natural killer cells attack the duct lining through a receptor called NKG2D, which recognises stress molecules displayed on injured cholangiocytes. Removing natural killer cells or blocking NKG2D preserved the continuity of the duct and the flow of bile, even though the virus and the inflammatory cytokines were still present.

In 2013, Tucker, Mack and colleagues showed that regulatory T cells in the newborn mouse liver are fewer, less activated and less effective. When regulatory T cells were transferred into infected newborn mice, survival rose to eighty-four percent, compared with 12.5 percent in controls. In the same year, Feldman and colleagues showed that mice lacking B cells were protected: disease-free survival was 76.8 percent, compared with 17.5 percent in normal mice, with no hyperbilirubinaemia, no duct obstruction and less type one cytokine production. If you picture these experiments as pairs of bars, each pair shows a short grey bar for the control animals and a long coloured bar for the animals in which one immune arm was removed or one brake was added. Each pair is a separate experiment, so the bars should not be compared across pairs, but together they make the point: interferon gamma, natural killer cells, B cells

and regulatory T cells each decide whether a virus-injured duct goes on to obliterate.

Part seven. Single cells in the human liver tell the same story.

Human data are, necessarily, descriptive. The most detailed picture comes from the 2020 study by Wang and colleagues in Cell, which profiled immune cells from atresia livers one cell at a time. They found four things. Macrophages had a blunted inflammatory programme, and Kupffer cells, the resident macrophages of the liver, were poor at clearing debris. Cytotoxic T cells were expanded, while a population of effector T cells and natural killer cells marked by a receptor called CX3CR1 was deficient. B cells were still being produced in the liver after birth, a process that normally stops, and these B cells showed defects in tolerance, with accumulation of IgG autoantibodies. In the rotavirus mouse model, depleting B cells or blocking antigen presentation reduced injury. And in a small pilot clinical study, the B-cell depleting antibody rituximab removed hepatic B cells and normalised the function of macrophages, Kupffer cells and T cells. That pilot was a mechanistic study, not an efficacy trial, and it does not tell us whether rituximab helps infants.

Other clues point the same way. In 2010, Lu, Mack and colleagues showed in the mouse model that antibodies against a cholangiocyte protein called alpha-enolase cross-react with viral proteins, a possible route from infection to autoimmunity, which immunologists call molecular mimicry. Muraji's group in Japan has long studied maternal microchimerism, the presence of a small number of the mother's cells in the infant. In a 2022 study of twelve infants, maternal DNA was found in the blood cells of seven, and a higher load was associated with a worse outcome. The bottom line is that the human data show what sustains the injury, not what started it.

Part eight. Testing the immune hypothesis: steroids answered the question.

If the immune response destroys the duct, then suppressing it after the Kasai should help. This idea was tested properly. The START trial, published by Bezerra and colleagues in 2014, randomised 140 infants to high-dose steroids or placebo within seventy-two hours of the Kasai. The steroid regimen was intravenous methylprednisolone four milligrams per kilogram per day for two weeks, then oral prednisolone two milligrams per kilogram per day for two weeks, then a taper, for thirteen weeks in total.

The primary end point was bile drainage at six months, defined as a total bilirubin below 1.5 milligrams per decilitre with the native liver. It was reached by 58.6 percent of infants on steroids and 48.6 percent on placebo. The adjusted difference was 8.7 percentage points, with a confidence interval running from minus 10.4 to plus 27.7, and the P value was 0.43, so the result was not significant. In infants operated before seventy days, the figures were 71.8 percent versus 56.8 percent, but that was a subgroup analysis. At twenty-four months, bile

drainage was 49.4 percent versus 39.8 percent, still not significant, and transplant-free survival was essentially identical, 58.7 percent versus 59.4 percent, with a hazard ratio of 1.0. The total number of infants with at least one serious adverse event was similar, eighty-one percent versus eighty percent, but serious events came earlier with steroids: by thirty days after surgery, 37.2 percent of the steroid group had had one, compared with 19 percent of the placebo group. If you picture the results as pairs of bars, the steroid bar is slightly longer for drainage, the two bars are the same length for survival, and the steroid bar is twice as long for early serious events.

The King's College Hospital trial of 2007, led by Davenport, used a lower dose in seventy-three infants. Transplant at twelve months was 26 percent with steroids and 35 percent with placebo, a difference that was not significant. Bilirubin was lower at one month in infants operated before seventy days, but the effect did not last to six months. The 2018 Cochrane review pooled both trials, 213 infants in all, and found no effect on death, transplant, jaundice clearance or adverse events, with low certainty of evidence.

The PRIME trial, a phase one and two study published by Mack and colleagues in 2019, gave intravenous immunoglobulin, one gram per kilogram, at three to five days, thirty days and sixty days after the Kasai to twenty-nine infants. Immunoglobulin was feasible and caused no serious adverse events attributable to it, but there was no improvement in bilirubin, and one-year native liver survival was 58.6 percent, compared with 70.5 percent in a historical placebo group. The difference was not significant, but it certainly did not suggest benefit.

The pattern is clear. Every adjuvant tested so far aimed at the last step of the cascade. Immune modulation after the duct is gone cannot rebuild it. The next generation of trials is aiming earlier, at protection of the lining cells.

Part nine. The toxin hypothesis: a plant in a dry dam.

The most elegant story in the field began with sheep. In 1964 and again in 1988, in New South Wales in Australia, newborn lambs were born with white stools, absent gallbladders and early cirrhosis, an illness that looked very much like biliary atresia. Several hundred lambs died on one farm in the 1988 outbreak. The ewes were healthy. The outbreaks followed droughts, when the ewes had grazed on land around a dam that is normally under water. The suspect was a plant that grows on the exposed lake floor, from the genus Dysphania, specifically Dysphania glomulifera and Dysphania littoralis.

In 2015, Lorent and colleagues, working with Michael Pack and Rebecca Wells in Philadelphia, split extracts of this plant into ninety-five fractions and tested each on zebrafish larvae. One fraction caused malformation of the gallbladder and extrahepatic bile ducts. The active molecule was named biliatresone. It is an isoflavonoid-related compound, chemically a one-two-diaryl-two-propenone, and it makes up about 1.84 percent of the dry weight of the

plant. The reactive part of the molecule is a structure that chemists call an enone, which binds to glutathione and to amino acids. Strikingly, it damages extrahepatic cholangiocytes while largely sparing hepatocytes and the ducts inside the liver, which is exactly the pattern of human atresia.

Biliatresone is proof of principle: a toxin from the environment can pass through a healthy mother and produce an atresia-like lesion in her offspring without harming her. But the caveat matters for the parents in front of you. No human exposure to biliatresone has ever been shown, and the plant has a limited range. The 2024 organoid authors themselves wrote that biliatresone itself is unlikely to be relevant to human atresia, and they suggested other chemicals with similar behaviour, such as microcystin-RR from algal blooms and aflatoxin B1, as candidates for study.

Part ten. Glutathione is the weak point.

Why does a toxin that binds glutathione hit the extrahepatic duct? In 2016, Waisbourd-Zinman and colleagues grew mouse extrahepatic cholangiocytes as small hollow spheres called spheroids, and exposed them to biliatresone. Glutathione fell by 43.6 percent within one hour, before any injury was visible. The spheroids then began to leak: the time for half of a dye called rhodamine to escape from the lumen was 4.4 hours with the toxin, compared with 11.4 hours with the solvent alone. Cilia were lost, and the lumens became obstructed within six to twelve hours. A chemical that depletes glutathione by a different route, buthionine sulfoximine, reproduced the injury. And the active form of N-acetylcysteine, which replenishes glutathione, prevented it, as did sulforaphane, which boosts glutathione production, whereas the inactive mirror-image form of N-acetylcysteine did not. Knocking down SOX17 alone obstructed the lumens without any change in glutathione, which places SOX17 downstream of glutathione loss. And extrahepatic duct tissue has less glutathione than liver tissue, which explains why the extrahepatic duct is the target.

The 2024 study by Yue and colleagues in Hong Kong repeated these experiments in human liver organoids. In control organoids, the proportion of ciliated cells rose to about fifty-six percent by day five; in organoids exposed to biliatresone, ciliated cells fell to zero by day three. Only 1.3 percent of exposed cholangiocytes produced a calcium signal in response to fluid flow, compared with twenty-eight percent of control cells, showing that the cilia had stopped working as flow sensors. The translation is simple. Glutathione depletion reproduces the lesion, and restoring glutathione rescues it. That is part of the rationale for giving N-acetylcysteine after the Kasai, which is now being tested in a phase two trial at Texas Children's Hospital. In that trial, intravenous N-acetylcysteine is given continuously for seven days starting within a day of surgery, and the primary end point is a normal serum bile acid level within twenty-four weeks.

Part eleven. Genetic susceptibility: risk genes, not a Mendelian disease.

Biliary atresia is not inherited in a simple pattern, but genes shift the risk. A 2010 genome-wide association study by Garcia-Barcelo and colleagues in Hong Kong, published in Human Molecular Genetics, compared two hundred Chinese infants with atresia with 481 controls and replicated the result in a second group. It found a susceptibility region on chromosome 10, at band q24.2, between the genes XPNPEP1 and ADD3, with a combined P value of about seven in a billion. ADD3 encodes a protein of the cell skeleton that helps hold epithelial junctions together. In 2013, Cui and colleagues in Philadelphia found that deletions removing one copy of the gene GPC1, on chromosome 2, were more common in sixty-one infants with atresia than in more than five thousand controls; knocking down the same gene in zebrafish caused biliary defects through the hedgehog signalling pathway. In 2019, Berauer and colleagues, working in the North American Childhood Liver Disease Research Network, known as ChiLDReN, sequenced sixty-seven children with the splenic malformation syndrome and found rare damaging variants in both copies of the ciliary gene PKD1L1 in five, and in one copy in three more. Other associations, such as EFEMP1 and ARF6, and rare variants in ciliary transport genes, have been reported in smaller studies.

The pattern is that common variants change risk modestly, while rare ciliary variants cluster in the most embryonic phenotype, the splenic malformation form. Twins tell the same story. In a 2020 report from eleven Chinese centres, all nineteen twin pairs were discordant for atresia, including identical pairs. Two-year native liver survival in the affected twins was 61.1 percent, and overall survival was 94.4 percent. Identical twins who share a genome and, often, a placenta, yet differ in whether they develop atresia, point to a chance event or a second hit rather than inheritance alone. Studies of cilia, such as the 2012 work by Chu, Russo and Wells, have shown that cholangiocyte cilia are abnormal in both syndromic and non-syndromic atresia, which suggests that a structurally vulnerable duct is common, and that a trigger then decides whether it fails.

Part twelve. From pathogenesis to bedside: injury you can measure.

The clearest translation of pathogenesis research into practice is MMP-7. In 2017, Lertudomphonwanit, Bezerra and colleagues screened 1,129 serum proteins in seventy infants and found seventy-six that differed between atresia and other causes of cholestasis. MMP-7 stood out. It is made by injured cholangiocytes, and in the mouse model it promotes the disease, so it is a sentinel of injury rather than a bystander. In combination with gamma-glutamyl transferase, known as GGT, it had an area under the receiver operating curve of 0.94 in validation, with a sensitivity of ninety-seven percent and a specificity of ninety-four percent. At a prevalence of about twenty-six percent among infants with conjugated hyperbilirubinaemia, that would give a positive predictive value of eighty-five percent and a negative predictive value of ninety-nine percent. Because MMP-7 reads out step three of the

cascade, epithelial injury, it rises early and stays high. The details of cut-offs and how to combine it with ultrasound belong to the next chapter, on diagnosis.

Part thirteen. What the guidelines say.

The 2017 joint NASPGHAN and ESPGHAN guideline states that the cause of biliary atresia is unknown, lists developmental and genetic, viral, toxic and immune mechanisms, and says that a role for cytomegalovirus remains unproven. On aetiology and timing, the current evidence is aligned with the guideline: a multi-hit model with no single cause, and an injury that begins before or soon after birth. On cytomegalovirus, the evidence differs in emphasis: it is not proven to be causal, but it defines a subgroup with worse outcomes and its prevalence rises with age at surgery. On steroids, the guideline does not support routine use, and the START trial and the Cochrane review are aligned with that. On immunoglobulin, rituximab and antivirals, the guideline is silent, which is a gap; the evidence shows no efficacy for immunoglobulin, only a mechanistic pilot for rituximab, and only non-randomised signals for antivirals. On toxins, the guideline mentions the hypothesis, and biliatresone has since been defined, without proof of human exposure. The one material difference concerns screening. The guideline asks clinicians to evaluate any infant who is still jaundiced at two to three weeks, but it predates the evidence that newborn direct bilirubin screening is feasible at scale, finds nearly every case, and brings the Kasai about nineteen days earlier.

Part fourteen. Where pathogenesis misleads at the cot side.

Several reasoning errors follow from half-understood mechanisms. The first is to think that a normal direct-to-total ratio in the first days of life means the jaundice is not cholestatic. In newborns with atresia, a ratio of 0.2 or less is common; use the absolute direct or conjugated value. The second is to think that a positive cytomegalovirus test means viral hepatitis. Cytomegalovirus coexists with atresia, infection never excludes it, and any infant with pale stools needs a cholangiogram. The third is to give steroids because the disease is immune. Two randomised trials were negative, and steroids brought serious events earlier. The fourth is to tell parents that the disease is genetic and that siblings are at high risk. Recurrence is rare and identical twins are discordant; genetic testing belongs mainly in syndromic cases. Two more errors are worth naming. A normal bilirubin at birth lowers the odds of atresia but does not remove the need to look at the stools. And when parents ask what they did in pregnancy, no maternal exposure has been proven in humans.

What to do differently in clinic.

Return to the mother and the questions from the beginning. You can tell her, honestly, that the cause of biliary atresia is not known, that the injury most likely began before her baby was born, and that nothing she did after birth caused it. No maternal exposure has been proven in humans. To the surgical team, you can say that routine steroids are not supported: the START

trial showed a non-significant ten-point difference in drainage, identical transplant-free survival and earlier serious adverse events, and the Cochrane review agrees. To the laboratory result, you can respond that a positive cytomegalovirus IgM does not change the operation; it identifies a subgroup with worse jaundice clearance, it should be recorded, and antivirals should be considered only within a protocol.

For a newborn on day two with a total bilirubin of nine and a direct bilirubin of 1.1 milligrams per decilitre, the ratio of about 0.12 does not reassure you; repeat a fractionated bilirubin by two weeks and check the stool colour. For the identical co-twin of an affected infant, the risk is low, and fractionated bilirubin and stool colour are more useful than genetic testing. For an infant with the splenic malformation syndrome and a situs anomaly, consider a ciliary gene panel including PKD1L1, obtain an echocardiogram, and explain that the disease began in early embryonic life.

So tomorrow in clinic, remember these numbers. The direct bilirubin of infants with atresia is about 1.4 milligrams per decilitre in the first days of life, compared with about 0.19 in healthy newborns. Newborn screening found every detected case, with specificity of 99.9 percent, and brought the Kasai from 56 to 36 days. Cytomegalovirus positivity rises with age at surgery, from none before thirty days to a third after seventy days. In the mouse model, removing interferon gamma, natural killer cell signalling or B cells, or adding regulatory T cells, prevents atresia. Steroids raised drainage from 48.6 to 58.6 percent, not significantly, and did not change survival, and immunoglobulin did not help. Biliatresone cuts glutathione by more than forty percent in an hour, and N-acetylcysteine prevents the injury in the dish. And identical twins are discordant. Above all, remember that the disease is already present at birth, so the absolute conjugated bilirubin and the colour of the stool, not the virus test or the bilirubin ratio, decide when to act.